

Constructing Decision Trees

Top-Down Approach

1. Start with the Root Node: Begin with entire dataset as the root.
2. Select the Best Feature: Evaluate all possible features using splitting criteria (eg. Gini Index or Information Gain) to determine the feature that best separates the data. This feature is called the decision node.
3. Split the Data: Divide the dataset into subsets based on the selected feature's values. Each subset will be represented by the branch of the tree.

4. Repeat for each subset: For each subset repeat the steps 2 & 3
 - If the subset is homogenous (all instances belong to the same class), create a leaf node and assign the class label.
 - If the stopping criteria (eg. maximum depth, minimum number of samples in leaf, or no further improvement) is reached, create a leaf node with the majority class label.
5. Continue until Complete and use the decision tree to assign labels to unknown samples.

Stopping Criteria	Description
> Maximum depth	Limits the height of the tree to prevent complexity
> Minimum samples per leaf	Ensures that the leaf nodes contain a minimum number of samples
> Minimum Information Gain (/ Gini Index)	Stops splitting if further splits do not significantly improve classification.
> Maximum number of leaf nodes	Limits the total number of leaf nodes in the tree.

eg. Consider a small dataset containing information about weather conditions and whether to play tennis or not. Construct a decision tree using top-down approach using information gain as the splitting criteria.

$\bar{z}$

↓ ↓ ↓ ↓ output (y)

	Outlook (3)	Temperature (3)	Humidity (2)	Windy (2)	Play Tennis (2)
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rain	Mild	High	False	Yes
5	Rain	Cool	Normal	False	Yes
6	Rain	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
10	Rain	Mild	Normal	False	Yes
11	Sunny	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rain	Mild	High	True	No

$$H(D) = - \left[\frac{5}{14} \times \log_2 \frac{5}{14} + \frac{9}{14} \times \log_2 \frac{9}{14} \right] = 0.9403$$

D: Y: 9, N: 5

				H	Ratio
Outlook:-	Sunny	Y: 2, N: 3	T: 5	0.971	5/14
	Overcast	Y: 4, N: 0	T: 4	0	4/14
	Rain	Y: 3, N: 2	T: 5	0.971	5/14

$$IG(D, Outlook) = 0.9403 - \left[\frac{5}{14} \times 0.971 + \frac{4}{14} \times 0 + \frac{5}{14} \times 0.971 \right]$$

$$\approx \boxed{0.2467}$$

				H	Ratio
Temperature:-	Hot	Y: 2, N: 2	T: 4	1	4/14
	Mild	Y: 4, N: 2	T: 6	0.9183	6/14
	Cool	Y: 3, N: 1	T: 4	0.8113	4/14

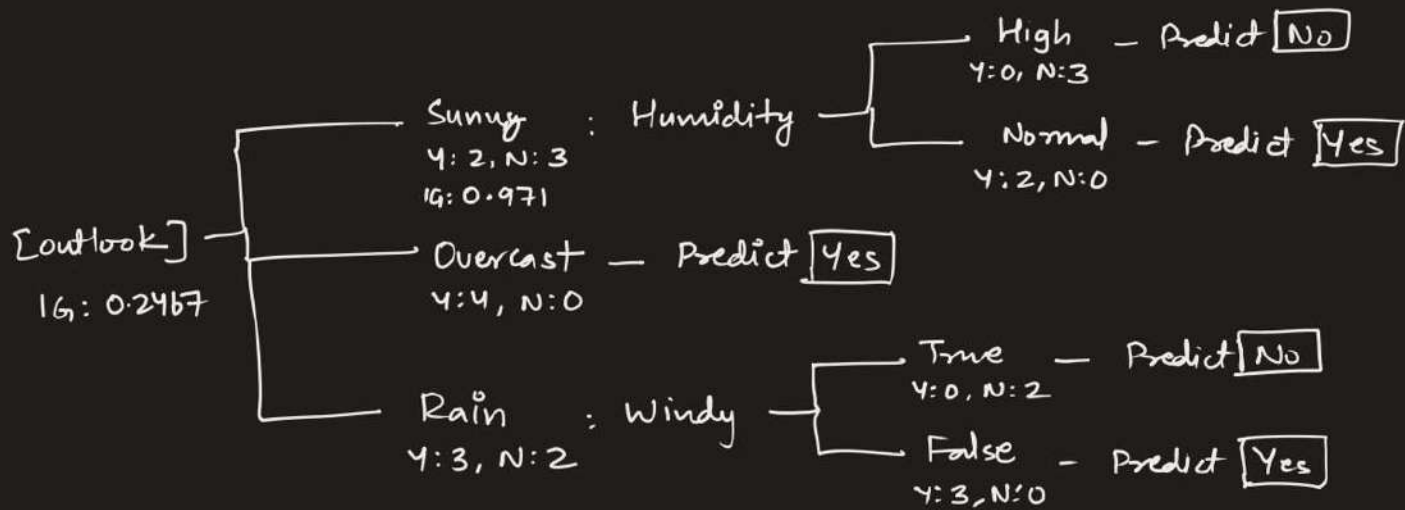
$$IG(D, Temperature) = 0.0292$$

				H	Ratio
Humidity:-	High	4:3, N:4	T:7	0.985	1/2
	Normal	4:6, N:1	T:7	0.592	1/2

$$IG(D, \text{Humidity}) = 0.1518$$

				H	Ratio
Windy:-	True	4:3, N:3	T:6	1	6/14
	False	4:6, N:2	T:8	0.8113	8/14

$$IG(D, \text{Windy}) = 0.0481$$



Sunny $H(D_S) = 0.971$

				H	Ratio
Temperature :	Hot	Y: 0	N: 2	0	2/5
	Mild	Y: 1	N: 1	1	2/5
	Cool	Y: 1	N: 0	0	1/5

$$IG(D_S, \text{Temperature}) = 0.971 - \left[\frac{2}{5} \times 0 + \frac{2}{5} \times 1 + \frac{1}{5} \times 0 \right]$$

$$= 0.571$$

		H	Ratio
Humidity :	High : 4:0, N:3	0	3/5
	Normal : 4:2, N:0	0	2/5

$$IG(D_5, \text{Humidity}) = 0.971 - 0 = \boxed{0.971}$$

Windy : ...

			H	Ratio
<u>Rain</u> $H(D_R) = 0.971$	Windy :	True 4:0, N:2	0	2/5
		False 4:3, N:0	0	3/5

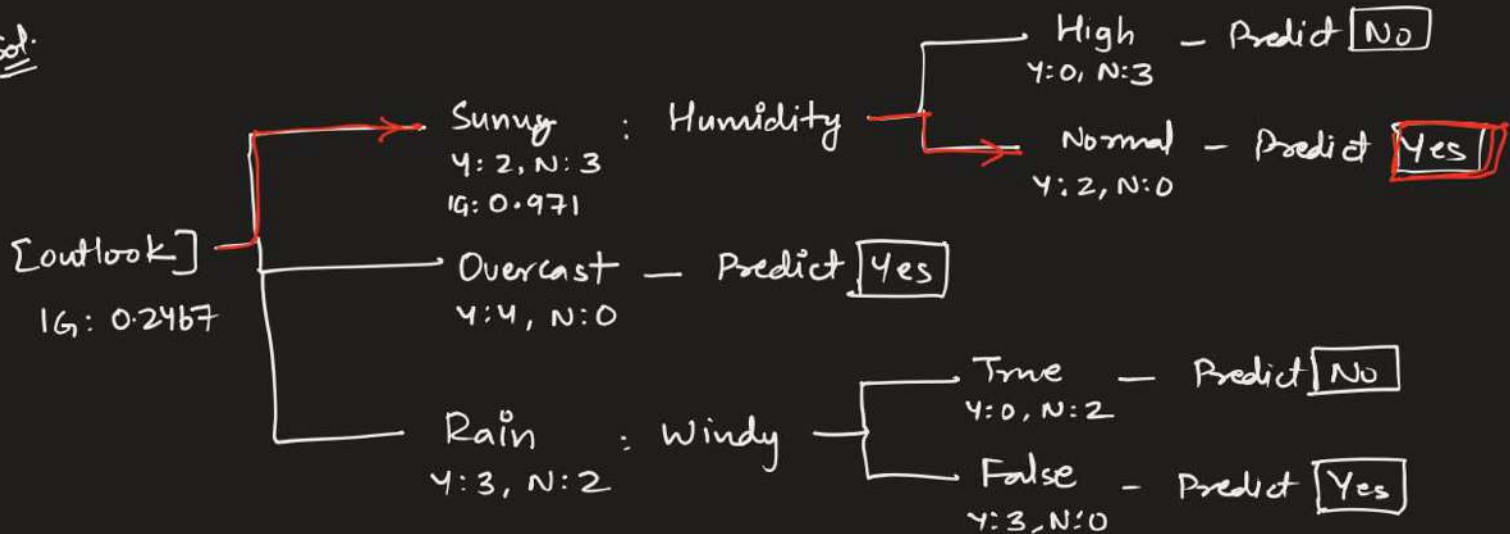
$$IG(D_2, \text{Windy}) = \boxed{0.971}$$

Temperature ...

Humidity ...

eg. (ii) classify a new point (Sunny, Mild, Normal, False) as Yes/No

Sol.



Follow the decision tree : Yes

How do we handle numerical data?

1. Identify Unique Values : Identify all unique values within the numerical feature in the dataset.
2. Calculate Potential Splits : For each unique value, consider it as a potential threshold for splitting the data into 2 subsets.
3. Calculate Information Gain : For each potential split, calculate the information gain to determine how much uncertainty is reduced when the dataset is split at that point.
4. Choose the best split : Select the point with maximum information gain.

eg. Consider a small dataset of animals with a numerical feature (weight) and a target variable indicating that the animal is a mammal or not.



Animal ID	Weight (kg)	Class
1	22	Yes
2	25	Yes
3	47	Yes
4	15	No
5	40	Yes
6	10	No
7	5	No
8	35	Yes

D: Y: 5, N: 3

$H(D)$: 0.954

Weight:

Threshold values	$\leq$ Threshold	$>$ Threshold	Entropy	IG
5	Y: 0, N: 1	Y: 5, N: 2	0.863	0.091
10	Y: 0, N: 2	Y: 5, N: 1	...	...
→ 15	Y: 0, N: 3	Y: 5, N: 0	0	<u>0.954</u>
22	Y: 1, N: 3	Y: 4, N: 0	...	...
25	Y: 2, N: 3	Y: 3, N: 0	...	...
35	Y: 3, N: 3	Y: 2, N: 0	...	...
40	Y: 4, N: 3	Y: 1, N: 0	...	...

